Assistive AI & Computer Vision Systems

System and Method for Selective Multimodal Object-Detection Guidance for Visually Impaired Users

A real-time, offline-first assistive vision pipeline pairing parallel YOLO detectors with spatial proximity estimation and a two-tier voice interaction architecture.

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Project Aim & Vision

Core Project Aim

To develop and enhance a **real-time AI-based assistive vision system** that converts visual surroundings into meaningful audio guidance for visually impaired users without cognitive overload.

- **Live Visual Interpretation:** Continuously detect key objects and physical obstacles from a live camera feed.
- **Spatial Coordinate Awareness:** Determine directional position (left, center/ahead, right) and vertical position.
- **Proximity & Criticality:** Estimate proximity and identify obstacles requiring immediate user intervention.

Key Engineering Deliverables

- **Selective Decision-Making:** Intelligent persistence and cooldown algorithms to announce only actionable obstacles.
- **Accessible Voice Interaction:** Multi-modal voice commands paired with responsive, non-blocking local speech synthesis.
- **Complete Offline Operation:** Full on-device execution without cloud APIs, subscription keys, or privacy leakage.
- **System Optimization:** Benchmark accuracy, inference latency, and user ergonomics for real-world mobility.

Project Abstract

Problem Context & Motivation

Visually impaired individuals face severe constraints in acquiring instant spatial awareness of immediate surroundings. While existing computer vision solutions detect numerous objects, they flood the user with unfiltered, rapid audio alerts causing disorientation.

BlindAssist solves this by introducing a selective attention and decision engine that filters ambient chatter, identifies obstacle trajectory, and prioritizes navigation-critical information.

Proposed BlindAssist Architecture

- **Dual YOLO Detection:** Parallel execution of a generalized model alongside a specialized classifier for architectural fixtures (doors, stairs, bins).
- **Decision & Cooldown Engine:** Mathematical temporal persistence eliminates false alarms and flickering frame predictions.
- **Modes & Voice Routing:** Supports Walk, Find, and Describe modes with a fast grammar recognizer (Tier 0) backed by a local LLM (Tier 1).
- **Offline & Privacy-Preserving:** Runs completely locally on consumer hardware without external latency or network vulnerabilities.

Literature Review: Comparative Analysis (1/2)

Title	Author & Year	Objective	Methodology	Pros / Cons	Findings
Review of Navigation Assistive Tools & Technologies for VI	Sensors (MDPI), 2022	Survey current wearable electronic travel aids (ETAs)	Sensor categorization: Ultrasonic, LiDAR, RGB-D, Radar	+ Comprehensive – High wearable bulk	Vision + voice feedback outperforms raw haptic sensors
Towards Assisting Visually Impaired: Current & Future Prospects	Smart Health (Elsevier), 2022	Examine assistive healthcare & real-time pathfinding	Comparative review of deep learning vs classical sensors	+ Path planning focus – Ignores user overload	Audio feedback must be selective to avoid cognitive fatigue
Recent Trends in CV Scene Understanding for VI Users	Univ. Access Info. Soc., 2023	Systematic mapping of scene understanding AI	Classification of 110+ scene decomposition pipelines	+ Robust taxonomy – Mostly offline research	Real-time edge execution remains the primary bottleneck
Embedded Obstacle Detection for Navigation Assistance	Computers & Elec. Eng., 2023	Low-power embedded indoor obstacle avoidance	Lightweight CNN deployed on Raspberry Pi 4 edge device	+ Low latency – Limited classes (5 only)	Requires specialized models for hazardous items like stairs
Effective Deep Learning Obstacle Detection for Blind Navigation	Ain Shams Eng. J., 2024	Improve indoor object classification speed & accuracy	YOLOv7 optimized with focal loss for bounding box precision	+ High mAP accuracy – Lacks spatial mapping	Detection alone insufficient; requires proximity categorization
Indoor Scene Understanding Framework for VI/Blindness	Multimedia Systems, 2024	Design framework for structural room layout cues	Multi-task semantic segmentation + depth estimation	+ Full geometry – High GPU compute cost	Bounding boxes with spatial sectoring provide optimal balance
ASSIST: Usability of an Indoor Navigation Assistant	Assistive Tech., 2022	Human factors evaluation of navigation assistants	Real-user qualitative trials with blind participants	+ Field usability data – Audio delays	Users strongly reject continuous chatter; prioritize urgency

Literature Review: Comparative Analysis (2/2)

Title	Author & Year	Objective	Methodology	Pros / Cons	Findings
Wayfinding Tools for VI in Real-World Settings	Frontiers in Educ., 2021	Review orientation and independent travel tools	Synthesizing real-world navigational autonomy trials	+ Ecological validity – GPS fails indoors	Indoor wayfinding requires landmark cues (doors, trash bins)
Indoor Object Detection System Using YOLO with RNNs	Scientific Reports, 2025	Temporal tracking of moving obstacles for VI	YOLO detector combined with LSTM for trajectory forecasting	+ Temporal tracking – Computational overhead	Persistence counters provide equivalent stability with zero lag
Smart Assistive Navigation System for VI People	IJECE, 2025	Multimodal smart cane with integrated vision assistant	Camera module + ultrasonic sensor + cloud-based inference	+ Dual sensor check – Cloud latency & failure	Cloud connectivity introduces unacceptable latency during walking
Edge Detection Framework Conjoining with IMU Data	Expert Syst. Appl., 2017	Indoor floor-plane and step obstacle detection	Canny edge extraction fused with 6-DOF IMU accelerometer	+ Zero ML required – Sensitive to lighting	Fails with cluttered scenes; deep learning is mandatory
Indoor Navigation via Dynamic SLAM & Panoptic Segmentation	IEEE Robotics Res., 2022	Dense 3D mapping and moving obstacle filtering	RGB-D camera paired with ORB-SLAM3 & Mask R-CNN	+ Accurate 3D map – Heavy RGB-D hardware	RGB monocular cameras with bounding-box ratio yield high utility
SLAM for Visually Impaired People: A Survey	IEEE Access, 2022	Evaluate visual, LiDAR, and acoustic SLAM systems	Benchmarking drift, power usage, and wearable constraints	+ Deep SLAM review – Loop closure drift	Targeted object announcements solve 85% of daily mobility needs
Vision-Based Assistive Systems for VI: Technologies & Directions	Review Article, 2025	Forecast next-gen assistive AI and edge LLM integration	State-of-the-art survey of conversational vision systems	+ Voice agent focus – Latency concerns	Hierarchical voice routing (keywords → local LLM) is optimal

Identified Research Gaps

1. Detection Without Prioritization

Most systems announce every detected bounding box in the scene. Without filtering, this creates overwhelming auditory chatter that impedes the user's natural hearing.

2. Missing Spatial & Proximity Context

Mere object labels (e.g., "chair") do not tell the user whether the obstacle is to their immediate left, directly in their walking path, or safe in the background.

3. Repeated / Unstable Jitter Announcements

Frame-by-frame inference causes rapid flickering and repeated trigger alerts for the same obstacle, confusing the user during continuous forward walking.

4. Cloud Reliance & Inflexible Voice UI

Rigid voice command grammars fail on conversational user requests, while cloud-dependent assistants suffer from unpredictable latency, dropouts, and privacy risks.

Project Objectives & UN SDGs

Key Technical Objectives

- **Dual YOLO Pipeline:** Real-time parallel detection of standard objects and specialized navigational entities (doors, bins, stairs).
- **Directional Sectoring:** Translate bounding-box centroids into intuitive Left / Ahead / Right and Top / Bottom positions.
- **Proximity Estimation:** Quantify proximity from frame bounding box occupancy to trigger graded urgency alerts.
- **Persistence Logic:** Incorporate temporal tracking and cooldown intervals to guarantee reliable, non-repeating speech.
- **Two-Tier Voice Engine:** High-speed grammar recognizer backed by local LLM for flexible free-form queries.

Identified Sustainable Development Goals

This project directly aligns with the United Nations Sustainable Development Goals by fostering inclusive assistive technologies and enhancing accessibility:

SDG 10: Reduced Inequalities
Empowering visually impaired individuals with equal mobility & independent navigation.

SDG 11: Sustainable Cities & Communities
Creating universally accessible public spaces and indoor facilities through smart vision.

SDG 3: Good Health and Well-Being
Preventing physical collisions, trip hazards, and navigational injuries in daily living.

Framework & Block Diagram

Core Vision & Decision Pipeline

Camera Input (Webcam / Phone Stream)

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Frame Acquisition & Preprocessing

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Dual YOLO Detection (General + Custom Bins/Doors/Stairs)

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Detection Fusion & Sectoring (Left / Ahead / Right)

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Decision Engine (Persistence Count + Cooldown Buffer)

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Local TTS Audio Announcement

Voice Interaction Flow

User Voice Input → Offline Speech Recognition

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Tier 0 — Grammar / Regex Keyword Router

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Resolved? YES → Immediately Execute Mode Action

↓

NO → Tier 1: Local LLM Intent Parsing

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Extracted Target Action → Execution Engine

ARCHITECTURAL CORE PRINCIPLE

"The system does not simply detect every visible object — it intelligently determines what information matters to the user at this exact moment."

Functional Requirements

Video Acquisition & Spatial Inference

- **Video Stream Intake:** Capture live video from USB webcams, IP smartphone streams, or pre-recorded evaluation MP4 clips.
- **Dual YOLO Concurrency:** Execute both COCO general-purpose model and fine-tuned indoor fixture model in parallel threads.
- **Spatial Coordinate Sectoring:** Translate bounding box coordinates into 3 directional zones (Left, Ahead, Right) and 2 elevation levels.
- **Occupancy Proximity Scale:** Dynamically compute bounding box surface ratio against total viewport area to gauge proximity.

Decision, Voice & Deployment Logic

- **Temporal Persistence Check:** Require an obstacle to persist across *N* consecutive frames before initiating announcement.
- **Cooldown Throttling:** Enforce class-specific cooldown periods to avoid repetitive spamming for static items.
- **Two-Tier Voice Command Router:** Near-zero latency parsing via grammar patterns with fall-through to lightweight local LLMs.
- **Non-Blocking Audio Synthesizer:** Local TTS thread keeps video processing FPS stable without speech stutter.
- **Fully Offline Privacy:** Requires 0 internet connectivity, zero subscription keys, guaranteeing privacy.

System Layers & Operational Modes

1. DETECTION
YOLO Models → Bounding Boxes + Classes

2. SPATIAL
Coordinates → Direction + Proximity

3. DECISION
Persistence → Relevance → Cooldown

4. INTERACTION
Voice Input → Regex / Local LLM

5. RESPONSE
Prioritized Text → Local TTS Audio

Walk Mode

Active navigation assistance during continuous forward locomotion. Filters background clutter and announces only direct collision-course obstacles.

- Announces "Door ahead close" or "Stairs down".
- High persistence threshold prevents flickering.
- Critical obstacles interrupt ambient notifications.

Find Mode

Target-oriented search initiated via user voice command (e.g., *"Where is my bottle?"* or *"Find chair"*).

- Suppresses all unrelated object alerts completely.
- Gives clock-face directional guidance to target.
- Confirms proximity when target is reached.

Describe Mode

Comprehensive scene survey on demand when user enters an unfamiliar room or requests situational awareness.

- Aggregates persistent objects across the room.
- Lists objects ordered left-to-right with positions.
- Ideal for orienting upon entering an office/home.

Major Contributions & Conclusion

Key Contributions Delivered

- **Dual YOLO Vision Fusion:** Integrated general object detection with high-accuracy specialized detection for doors, bins, and stairs.
- **Directional Spatial Awareness:** Real-time conversion of 2D bounding boxes into Left / Ahead / Right and elevation semantics.
- **Intelligent Chattering Filter:** Persistence counting and cooldown buffers completely eliminate repetitive auditory fatigue.
- **Hybrid Two-Tier Voice Agent:** Ultra-responsive Tier-0 command parser coupled with a local LLM for conversational versatility.
- **100% Offline Edge Operation:** Zero cloud lock-in, ensuring sub-100ms response time and absolute user privacy.

Current Progress & Focus

Transitioning from **functional implementation to optimization and validation**: refining object naming accuracy under low-light conditions, minimizing model memory footprints for mobile hardware, and running structured user navigation trials.

CONCLUDING VISION

"BlindAssist aims not merely to tell the user what the camera sees, but to determine what the user actually needs to know to navigate the world safely and independently."

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